EXP 801 DS Wheat Heat Drought

— Ed Chaplin (USyd), *Apr–June 2025* —

Li-600 STOMATAL CONDUCTANCE MODEL PARAMETER ESTIMATION - LIGHT ON TO LIGHT OFF (DECREASE)

- Using the final model fitted to the data, obtain the parameters of interest
- These include:
- k_d, being the time for gs to decrease to Gsmin
- lambda, being the lag in the time of gs to a step decrease in light
- g0, being the starting value of the gs (intercept/upper asymptote)
- g1, being the lower asymptote of the gs
- Slmax, a derived trait from McAusland paper
-

Load the libraries

```
library(knitr)
#knitr::spin("Exp801.porometer.stomatalCondParamEstimation_d.v1.r")
library(ggplot2)
opts_chunk$set("tidy" = FALSE, "dev" = "png", "dpi" = 300,
               "fig.width" = 7.5, "fig.height" = 10, "comment"=NA, fig.path = "figures/increase/")
library(dae)
library(dplyr)
library(brms) ##package used to fit bayesian regress.off model using Stan
library(tidyr)
library(emmeans)
library(tidybayes)
```

Set up .offstants and files

```
opt.offs(width=90)
```

```
Error in `opt.offs()`:
! could not find function "opt.offs"
```

```
load("../data/porometer.anal_d.v1.Rdata")
saveFile <- "../data/stomatal.offdParamEstimat.off_porometer_d.v1.Rdata"
```

Have a look at the final model summary and store fixed effects

```
summary(fit_off_final)
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k) + 1))
         g0 ~ 1 + Line * Device_Watering * Room_Temperature + (1 | CEBRoom:CEBSide:CEBLocation + CEBRoo
         g1 ~ 1 + Line + Device_Watering + Line:Device_Watering + (1 | CEBRoom:CEBSide:CEBLocation + CE
         k ~ 1 + Room_Temperature
         lambda ~ 1 + Device_Watering
   Data: li600.off.dat (Number of observations: 1959)
  Draws: 6 chains, each with iter = 10000; warmup = 6000; thin = 1;
         total post-warmup draws = 24000
```

Multilevel Hyperparameters:
~CEBRoom:CEBSide:CEBLocation (Number of levels: 8)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(g0_Intercept)	0.04	0.03	0.00	0.11	1.00	11691	14175
sd(g1_Intercept)	0.00	0.00	0.00	0.01	1.00	9771	14474

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit (Number of levels: 16)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(g0_Intercept)	0.06	0.02	0.01	0.11	1.00	6481	6006
sd(g1_Intercept)	0.01	0.00	0.01	0.02	1.00	13702	17666

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell (Number of levels: 159)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(g0_Intercept)	0.14	0.01	0.12	0.16	1.00	8078	14856
sd(g1_Intercept)	0.01	0.00	0.01	0.01	1.00	11004	15780

Regression Coefficients:

	Estimate	Est.E
g0_Intercept	0.36	
g0_LinePBI17N002M0CM0NM0NM010NM16N	-0.17	
g0_LinePBI17N006M0NM0NM89N	0.06	
g0_LinePBI17N007M0CM0NM0NM010NM8N	0.02	
g0_LinePBI17N013M0CM0NM0NM010NM8N	0.02	
g0_LinePBI17N016M0CM0NM0NM010NM27N	-0.20	
g0_LinePBI19N007M0NM3N	0.13	
g0_LinePBIC15034M0CM24NM010NM1NM0N	0.04	
g0_LineROCKSTAR	0.05	
g0_LineSUNCHASER	0.12	
g0_Device_WateringD2_Limited	-0.31	
g0_Room_TemperatureCER8_Ambient	0.17	
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	0.22	
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.03	
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	-0.00	
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.08	
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	0.21	
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited	0.09	
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.15	
g0_LineROCKSTAR:Device_WateringD2_Limited	0.15	
g0_LineSUNCHASER:Device_WateringD2_Limited	0.03	
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	-0.20	
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	-0.18	
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	-0.29	
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	-0.16	
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	-0.14	
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	-0.12	
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	-0.15	
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient	-0.19	
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient	-0.17	
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.09	
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.07	
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.06	
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.16	
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.12	
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.16	

```
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                -0.10
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient    -0.05
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                         0.19
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                        0.09
g1_Intercept                                                                                  0.00
g1_LinePBI17N002M0CM0NM0NM010NM16N                                                           -0.00
g1_LinePBI17N006M0NM0NM89N                                                                   -0.00
g1_LinePBI17N007M0CM0NM0NM010NM8N                                                             0.01
g1_LinePBI17N013M0CM0NM0NM010NM8N                                                            -0.00
g1_LinePBI17N016M0CM0NM0NM010NM27N                                                           -0.00
g1_LinePBI19N007M0NM3N                                                                        0.00
g1_LinePBIC15034M0CM24NM010NM1NM0N                                                           -0.00
g1_LineROCKSTAR                                                                               0.00
g1_LineSUNCHASER                                                                              0.00
g1_Device_WateringD2_Limited                                                                  0.00
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited                                 -0.00
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited                                         -0.00
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                  -0.01
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                   0.00
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited                                 -0.01
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited                                              -0.01
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited                                 -0.00
g1_LineROCKSTAR:Device_WateringD2_Limited                                                     -0.01
g1_LineSUNCHASER:Device_WateringD2_Limited                                                    -0.01
k_Intercept                                                                                   2.62
k_Room_TemperatureCER8_Ambient                                                                1.97
lambda_Intercept                                                                              0.02
lambda_Device_WateringD2_Limited                                                              0.03

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.02      0.00     0.02     0.02 1.00    28903    19031

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Build a dataframe of model parameters

```
li600.d.param <- expand.grid(Room_Temperature = levels(li600.off.dat$Room_Temperature),
                             Device_Watering = levels(li600.off.dat$Device_Watering),
                             Line = levels(li600.off.dat$Line))
```

Use the emmeans package to get out the expected parameter estimates, a.offg with HPD (credible intervals)

```
g0.emmeans <- as.data.frame(emmeans(fit_off_final,
                                    ~ Room_Temperature * Device_Watering * Line,
                                    nlpar = "g0"))
g0.emmeans <- g0.emmeans %>% rename_with(~paste0("g0_", .x),
                                         -all_of(c("Room_Temperature",
                                                   "Device_Watering",
                                                   "Line")))
```

```
g0.emmeans <- within(g0.emmeans, {
  g0_tabquant <- paste0("'", round(g0_emmean, 3), "\n[",
                        round(g0_lower.HPD, 3), ", ",
                        round(g0_upper.HPD, 3), "]'")

})
li600.d.param <- left_join(li600.d.param, g0.emmeans)
```

```
Joining with `by = join_by(Room_Temperature, Device_Watering, Line)`
```

```
g1.emmeans <- as.data.frame(emmeans(fit_off_final,
                                    ~ Device_Watering * Line,
                                    nlpar = "g1"))
g1.emmeans <- g1.emmeans %>% rename_with(~paste0("g1_", .x),
                                         -all_of(c("Device_Watering",
                                                   "Line")))
g1.emmeans <- within(g1.emmeans, {
  g1_tabquant <- paste0("'", round(g1_emmean, 3), "\n[",
                        round(g1_lower.HPD, 3), ", ",
                        round(g1_upper.HPD, 3), "]'")

})
li600.d.param <- left_join(li600.d.param, g1.emmeans)
```

```
Joining with `by = join_by(Device_Watering, Line)`
```

```
k.emmeans <- as.data.frame(emmeans(fit_off_final,
                                   ~ Room_Temperature,
                                   nlpar = "k"))
k.emmeans <- k.emmeans %>% rename_with(~paste0("k_", .x),
                                       -all_of(c("Room_Temperature")))
k.emmeans <- within(k.emmeans, {
  k_tabquant <- paste0("'", round(k_emmean, 3), "\n[",
                       round(k_lower.HPD, 3), ", ",
                       round(k_upper.HPD, 3), "]'")

})
li600.d.param <- left_join(li600.d.param, k.emmeans)
```

```
Joining with `by = join_by(Room_Temperature)`
```

```
lambda.emmeans <- as.data.frame(emmeans(fit_off_final,
                                  ~ Device_Watering,
                                  nlpar = "lambda"))
lambda.emmeans <- lambda.emmeans %>% rename_with(~paste0("lambda_", .x),
                                      -all_of(c("Device_Watering")))
lambda.emmeans <- within(lambda.emmeans, {
  lambda_tabquant <- paste0("'", round(lambda.emmean, 3), "\n[",
                      round(lambda.lower.HPD, 3), ", ",
                      round(lambda.upper.HPD, 3), "]'")

})
```

```
Error:
! object 'lambda.emmean' not found
```

```
li600.d.param <- left_join(li600.d.param, lambda.emmeans)
```

```
Joining with `by = join_by(Device_Watering)`
```

Using these parameters, calculate Slmax

- Slmax looks to be reported in the units mmol mˆ-2 sˆ-2
- This involves multiplying g1 and g0 by 1000 and also k by 60
- I can calculate this using the posterior draws for each parameter and then combine them together

```
g0_draws <- emmeans(fit_off_final,
                    ~ Room_Temperature * Device_Watering * Line,
                    nlpar = "g0") %>%
  gather_emmeans_draws()

g1_draws <- emmeans(fit_off_final,
                    ~ Device_Watering * Line,
                    nlpar = "g1") %>%
  gather_emmeans_draws()

k_draws <- emmeans(fit_off_final,
                   ~ Room_Temperature,
                   nlpar = "k") %>%
  gather_emmeans_draws()

combined_draws <- g0_draws %>%
  rename(g0 = .value) %>%
  left_join(g1_draws %>% rename(g1 = .value),
            by = c(".draw", "Device_Watering", "Line")) %>%
  left_join(k_draws %>% rename(k = .value),
            by = c(".draw", "Room_Temperature"))

combined_draws <- combined_draws %>%
  mutate(Slmax = (g1*1000 - g0*1000) / (exp(1) * k*60))

summary_slopes <- combined_draws %>%
  group_by(Room_Temperature, Device_Watering, Line) %>%
  median_hdi(Slmax) %>%
  rename(Slmax_emmean = Slmax) %>%
  rename(Slmax_lower.HPD = .lower) %>%
  rename(Slmax_upper.HPD = .upper)

summary_slopes <- within(summary_slopes, {
  Slmax_tabquant <- paste0("'",round(Slmax_emmean, 3), "\n[",
                           round(Slmax_lower.HPD, 3), ", ",
                           round(Slmax_upper.HPD, 3), "]'")

})

li600.d.param <- left_join(li600.d.param,
                           summary_slopes %>% select(Room_Temperature,
                                                     Device_Watering,
                                                     Line,
                                                     Slmax_emmean,
```

```
                                                    Slmax_lower.HPD,
                                                    Slmax_upper.HPD,
                                                    Slmax_tabquant))
```

```
Joining with `by = join_by(Room_Temperature, Device_Watering, Line)`
```

```
li600.d.param
```

```
   Room_Temperature Device_Watering                      Line  g0_emmean g0_lower.HPD g0_upper.HPD
1         CER7_Heat    D1_Unlimited PBI16C009-0C-0N-0N-010N-14N 0.35504387  0.223190153    0.4875488
2      CER8_Ambient    D1_Unlimited PBI16C009-0C-0N-0N-010N-14N 0.53057635  0.387970458    0.6692195
3         CER7_Heat      D2_Limited PBI16C009-0C-0N-0N-010N-14N 0.04495835 -0.102484021    0.1874464
4      CER8_Ambient      D2_Limited PBI16C009-0C-0N-0N-010N-14N 0.12736297 -0.023201437    0.2714393
5         CER7_Heat    D1_Unlimited PBI17N002-0C-0N-0N-010N-16N 0.18433032  0.032223940    0.3429379
6      CER8_Ambient    D1_Unlimited PBI17N002-0C-0N-0N-010N-16N 0.15995677  0.007042050    0.3135489
7         CER7_Heat      D2_Limited PBI17N002-0C-0N-0N-010N-16N 0.09554651 -0.061785945    0.2463861
8      CER8_Ambient      D2_Limited PBI17N002-0C-0N-0N-010N-16N 0.05003035 -0.100548690    0.2073912
9         CER7_Heat    D1_Unlimited         PBI17N006-0N-0N-89N 0.41660011  0.264659892    0.5674929
10     CER8_Ambient    D1_Unlimited         PBI17N006-0N-0N-89N 0.40830001  0.251336357    0.5570386
11        CER7_Heat      D2_Limited         PBI17N006-0N-0N-89N 0.13764445 -0.013486463    0.2961097
12     CER8_Ambient      D2_Limited         PBI17N006-0N-0N-89N 0.10136644 -0.056595466    0.2529579
13        CER7_Heat    D1_Unlimited  PBI17N007-0C-0N-0N-010N-8N 0.37970102  0.229958524    0.5329856
14     CER8_Ambient    D1_Unlimited  PBI17N007-0C-0N-0N-010N-8N 0.26744964  0.118268863    0.4255467
15        CER7_Heat      D2_Limited  PBI17N007-0C-0N-0N-010N-8N 0.06696552 -0.088160484    0.2244000
16     CER8_Ambient      D2_Limited  PBI17N007-0C-0N-0N-010N-8N 0.01848235 -0.137019372    0.1689897
17        CER7_Heat    D1_Unlimited  PBI17N013-0C-0N-0N-010N-8N 0.38050194  0.224512634    0.5303086
18     CER8_Ambient    D1_Unlimited  PBI17N013-0C-0N-0N-010N-8N 0.40035052  0.249692370    0.5535988
19        CER7_Heat      D2_Limited  PBI17N013-0C-0N-0N-010N-8N 0.15174256 -0.018206969    0.3296940
20     CER8_Ambient      D2_Limited  PBI17N013-0C-0N-0N-010N-8N 0.19524012  0.040076195    0.3512786
21        CER7_Heat    D1_Unlimited PBI17N016-0C-0N-0N-010N-27N 0.15945749  0.007597480    0.3085600
22     CER8_Ambient    D1_Unlimited PBI17N016-0C-0N-0N-010N-27N 0.19533789  0.037235064    0.3459520
23        CER7_Heat      D2_Limited PBI17N016-0C-0N-0N-010N-27N 0.06356841 -0.092069626    0.2213411
24     CER8_Ambient      D2_Limited PBI17N016-0C-0N-0N-010N-27N 0.16779927  0.008047408    0.3199940
25        CER7_Heat    D1_Unlimited             PBI19N007-0N-3N 0.48483313  0.334777500    0.6366735
26     CER8_Ambient    D1_Unlimited             PBI19N007-0N-3N 0.54381455  0.390866194    0.6957003
27        CER7_Heat      D2_Limited             PBI19N007-0N-3N 0.26318417  0.110955720    0.4218045
28     CER8_Ambient      D2_Limited             PBI19N007-0N-3N 0.12608999 -0.029738993    0.2808076
29        CER7_Heat    D1_Unlimited PBIC15034-0C-24N-010N-1N-0N 0.39512486  0.244535075    0.5536881
30     CER8_Ambient    D1_Unlimited PBIC15034-0C-24N-010N-1N-0N 0.41611186  0.262706998    0.5719578
31        CER7_Heat      D2_Limited PBIC15034-0C-24N-010N-1N-0N 0.23729447  0.078996619    0.3916621
32     CER8_Ambient      D2_Limited PBIC15034-0C-24N-010N-1N-0N 0.11663296 -0.041826661    0.2701649
33        CER7_Heat    D1_Unlimited                   ROCKSTAR 0.40759140  0.258949683    0.5606586
34     CER8_Ambient    D1_Unlimited                   ROCKSTAR 0.38888940  0.235814290    0.5436577
35        CER7_Heat      D2_Limited                   ROCKSTAR 0.25007818  0.096043451    0.4023250
36     CER8_Ambient      D2_Limited                   ROCKSTAR 0.32576657  0.170348427    0.4820663
37        CER7_Heat    D1_Unlimited                  SUNCHASER 0.47747022  0.320942564    0.6276430
38     CER8_Ambient    D1_Unlimited                  SUNCHASER 0.48089341  0.326268944    0.6349031
39        CER7_Heat      D2_Limited                  SUNCHASER 0.19232587  0.036263572    0.3547791
40     CER8_Ambient      D2_Limited                  SUNCHASER 0.19279746  0.041880604    0.3541100
                 g1_tabquant k_emmean k_lower.HPD k_upper.HPD             k_tabquant lambda_emmean lam
1    '0.001\n[-0.01, 0.011]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.01763447
2    '0.001\n[-0.01, 0.011]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.01763447
3   '0.006\n[-0.005, 0.016]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.04680341
4   '0.006\n[-0.005, 0.016]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.04680341
5  '-0.002\n[-0.013, 0.009]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.01763447
```

```
6  '-0.002\n[-0.013, 0.009]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.01763447
7   '-0.001\n[-0.012, 0.01]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.04680341
8   '-0.001\n[-0.012, 0.01]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.04680341
9  '-0.002\n[-0.013, 0.009]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.01763447
10 '-0.002\n[-0.013, 0.009]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.01763447
11   '0.002\n[-0.01, 0.013]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.04680341
12   '0.002\n[-0.01, 0.013]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.04680341
13  '0.006\n[-0.005, 0.017]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.01763447
14  '0.006\n[-0.005, 0.017]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.01763447
15  '0.003\n[-0.009, 0.014]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.04680341
16  '0.003\n[-0.009, 0.014]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.04680341
17 '-0.003\n[-0.014, 0.008]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.01763447
18 '-0.003\n[-0.014, 0.008]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.01763447
19  '0.002\n[-0.009, 0.014]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.04680341
20  '0.002\n[-0.009, 0.014]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.04680341
21 '-0.002\n[-0.013, 0.009]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.01763447
22 '-0.002\n[-0.013, 0.009]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.01763447
23 '-0.002\n[-0.014, 0.009]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.04680341
24 '-0.002\n[-0.014, 0.009]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.04680341
25  '0.003\n[-0.008, 0.014]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.01763447
26  '0.003\n[-0.008, 0.014]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.01763447
27  '-0.002\n[-0.013, 0.01]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.04680341
28  '-0.002\n[-0.013, 0.01]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.04680341
29 '-0.004\n[-0.015, 0.007]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.01763447
30 '-0.004\n[-0.015, 0.007]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.01763447
31  '-0.001\n[-0.013, 0.01]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.04680341
32  '-0.001\n[-0.013, 0.01]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.04680341
33   '0.001\n[-0.01, 0.012]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.01763447
34   '0.001\n[-0.01, 0.012]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.01763447
35  '-0.001\n[-0.013, 0.01]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.04680341
36  '-0.001\n[-0.013, 0.01]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.04680341
37  '0.002\n[-0.009, 0.013]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.01763447
38  '0.002\n[-0.009, 0.013]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.01763447
39 '-0.003\n[-0.015, 0.008]' 2.615539    2.538767    2.695345 '2.616\n[2.539, 2.695]'    0.04680341
40 '-0.003\n[-0.015, 0.008]' 4.587102    4.476275    4.693512 '4.587\n[4.476, 4.694]'    0.04680341
   Slmax_upper.HPD             Slmax_tabquant
1      -0.51642760 '-0.831\n[-1.144, -0.516]'
2      -0.51697670 '-0.708\n[-0.898, -0.517]'
3       0.24704882  '-0.092\n[-0.436, 0.247]'
4       0.03570141  '-0.163\n[-0.359, 0.036]'
5      -0.07051462 '-0.436\n[-0.802, -0.071]'
6      -0.01492542 '-0.216\n[-0.426, -0.015]'
7       0.14064003  '-0.226\n[-0.586, 0.141]'
8       0.13640023  '-0.068\n[-0.275, 0.136]'
9      -0.62641438 '-0.981\n[-1.344, -0.626]'
10     -0.34532991 '-0.548\n[-0.756, -0.345]'
11      0.04053865   '-0.319\n[-0.69, 0.041]'
12      0.07482503  '-0.133\n[-0.342, 0.075]'
13     -0.52064247 '-0.877\n[-1.237, -0.521]'
14     -0.14804415 '-0.349\n[-0.562, -0.148]'
15      0.22060557  '-0.151\n[-0.516, 0.221]'
16      0.18473373  '-0.021\n[-0.226, 0.185]'
17     -0.53904857  '-0.898\n[-1.26, -0.539]'
18     -0.33475711  '-0.54\n[-0.743, -0.335]'
```

```
19      0.05764602  '-0.351\n[-0.761, 0.058]'
20     -0.05263493 '-0.258\n[-0.471, -0.053]'
21     -0.02173523 '-0.379\n[-0.734, -0.022]'
22     -0.06094785 '-0.264\n[-0.476, -0.061]'
23      0.20622966  '-0.154\n[-0.531, 0.206]'
24     -0.01884647 '-0.227\n[-0.438, -0.019]'
25     -0.77833537 '-1.129\n[-1.492, -0.778]'
26     -0.51389451 '-0.722\n[-0.926, -0.514]'
27     -0.25081685 '-0.621\n[-0.985, -0.251]'
28      0.03412901  '-0.171\n[-0.383, 0.034]'
29     -0.57675825 '-0.935\n[-1.311, -0.577]'
30     -0.35330939  '-0.561\n[-0.77, -0.353]'
31     -0.19582710 '-0.559\n[-0.934, -0.196]'
32      0.05247397  '-0.157\n[-0.365, 0.052]'
33     -0.60959417  '-0.953\n[-1.326, -0.61]'
34     -0.31408474 '-0.518\n[-0.728, -0.314]'
35     -0.21970749  '-0.589\n[-0.943, -0.22]'
36     -0.22803571 '-0.438\n[-0.648, -0.228]'
37     -0.74530851 '-1.113\n[-1.476, -0.745]'
38     -0.43222093  '-0.64\n[-0.848, -0.432]'
39     -0.07518326 '-0.459\n[-0.828, -0.075]'
40     -0.05571012 '-0.262\n[-0.474, -0.056]'
```

Format this in a nicer way for export to Excel

Keep the columns in the same order as Li6800 results and order first by parameter and then Line

```
li600.d.export <- li600.d.param %>% select(Room_Temperature, Device_Watering, Line,
                                           grep("tabquant", colnames(.))) %>%
  pivot_longer(cols = 4:ncol(.),
               names_to = "params",
               values_to = "values") %>%
  pivot_wider(names_from = c("Room_Temperature", "Device_Watering"),
              values_from = "values")
li600.d.export$params <- factor(li600.d.export$params, levels = c("g0_tabquant",
                                                                  "g1_tabquant",
                                                                  "k_tabquant",
                                                                  "lambda_tabquant",
                                                                  "Slmax_tabquant"))
li600.d.export <- li600.d.export %>% arrange(params, Line) %>%
  select("params", "Line", "CER7_Heat_D2_Limited", "CER8_Ambient_D2_Limited",
         "CER7_Heat_D1_Unlimited", "CER8_Ambient_D1_Unlimited")
li600.d.export
```

```
# A tibble: 40 × 6
   params      Line                          CER7_Heat_D2_Limited       CER8_Ambient_D2_Limited    CER7_H
   <fct>       <fct>                         <chr>                      <chr>                      <chr>
 1 g0_tabquant PBI16C009-0C-0N-0N-010N-14N "'0.045\n[-0.102, 0.187]'" "'0.127\n[-0.023, 0.271]'" "'0.35
 2 g0_tabquant PBI17N002-0C-0N-0N-010N-16N "'0.096\n[-0.062, 0.246]'" "'0.05\n[-0.101, 0.207]'"  "'0.18
 3 g0_tabquant PBI17N006-0N-0N-89N         "'0.138\n[-0.013, 0.296]'" "'0.101\n[-0.057, 0.253]'" "'0.41
 4 g0_tabquant PBI17N007-0C-0N-0N-010N-8N  "'0.067\n[-0.088, 0.224]'" "'0.018\n[-0.137, 0.169]'" "'0.38
 5 g0_tabquant PBI17N013-0C-0N-0N-010N-8N  "'0.152\n[-0.018, 0.33]'"  "'0.195\n[0.04, 0.351]'"   "'0.38
 6 g0_tabquant PBI17N016-0C-0N-0N-010N-27N "'0.064\n[-0.092, 0.221]'" "'0.168\n[0.008, 0.32]'"   "'0.15
 7 g0_tabquant PBI19N007-0N-3N             "'0.263\n[0.111, 0.422]'"  "'0.126\n[-0.03, 0.281]'"  "'0.48
 8 g0_tabquant PBIC15034-0C-24N-010N-1N-0N "'0.237\n[0.079, 0.392]'"  "'0.117\n[-0.042, 0.27]'"  "'0.39
```

```
 9 g0_tabquant ROCKSTAR                    "'0.25\n[0.096, 0.402]'"   "'0.326\n[0.17, 0.482]'"   "'0.408
10 g0_tabquant SUNCHASER                   "'0.192\n[0.036, 0.355]'"  "'0.193\n[0.042, 0.354]'"  "'0.47
#  30 more rows
```

Save data and export

```
save.image(saveFile)
write.csv(li600.d.export, "../data/li600.d.param.export.csv", row.names = FALSE,
          quote = TRUE)
```